EQUALITY OF THE SCOTT RANK OF A STRUCTURE AND OF THE SCOTT RANK OF ITS TREE OF TUPLES

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Editorial note. The retained excerpt at source lines 45-53 ends in the middle of the article's sentence 'While the latter is not an ...'; the sentence continues at line 54, which points at the invariant table and is deliberately not extracted. The three rank definitions inside the excerpt are complete.

Editorial note. The retained environments print with this package's own numbering, while the article numbers them within its sections; the original labels are preserved in the span metadata and no number is asserted here.

Editorial note. The article's translation lemma at source lines 234-286 is a source-local support result and is delivered with its complete proof; it is not a newly counted target.

Editorial note. No retained line contains a citation, a figure or an included file, so the package declares neither a bibliography nor an asset.

Editorial scope. One original exact result: the equality of the Scott rank of a structure and of its tree of tuples, together with the article's own complete translation lemma on which the proof rests.

Allowed standard prerequisites (editorial).

- Infinitary formulas and back-and-forth relations. The complexity classes Pi^in_alpha and Sigma^in_alpha of L_{omega_1,omega} formulas, the back-and-forth relation =_alpha, and Scott rank as the least ordinal whose Sigma^in formulas define all tuple orbits; all three ranks are defined in the retained context excerpt.
- Labeled trees and the tree-of-tuples construction. The vocabulary of labeled trees, the finite atomic diagram coding used as a label, and the replication of every branch; the article's own translation lemma (source lines 234-286) is delivered with its complete proof and is not assumed.

In this article we give a comprehensive Scott analysis of the class of linear orderings using three definitions of rank. Given a structure $\mathcal{A}$ the *(parameterless) Scott rank* $SR(\mathcal{A})$, is the least α, such that the automorphism orbits of all tuples in $\mathcal{A}$ are $\Sigma^{\text{in}}_\alpha$ definable; the *parameterized Scott rank* $pSR(\mathcal{A})$, is the least ordinal α such that there is a parameter $\bar{p} \in A^{<\omega}$ such that $SR((\mathcal{A}, \bar{p})) = \alpha$; and the *Scott sentence complexity* of $\mathcal{A}$, $SSC(\mathcal{A})$ is the least of the complexities Π^{in}_α, $\Sigma^{\text{in}}_\alpha$, d-$\Sigma^{\text{in}}_\alpha$ such that $\mathcal{A}$ has a Scott sentence of one of these complexities. While the latter is not an

0.1. **The tree of tuples and Scott sentence complexity.**

Definition 0.1. A *replicated labeled tree* consists of a tree $(T, \succeq)$ with a parent function and labeling function $l : T \to \omega$ that satisfies that for every $\sigma \in T$ with parent τ, there exist infinitely many children $\tilde{\sigma}$ of τ such that $\mathcal{T}_\sigma \cong \mathcal{T}_{\tilde{\sigma}}$.

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Definition 0.2. Given a structure $\mathcal{A}$ let $T_{\mathcal{A}}$ be the labeled tree consisting of all tuples from $\mathcal{A}$ ordered by inclusion where each tuple $\bar{a}$ is labeled by a natural number coding its finite atomic diagram $D_{\mathcal{A}}(\bar{a})$ and the length of $\bar{a}$. The *tree of tuples* $\mathcal{T}_{\mathcal{A}}$ is obtained by replicating every branch in $T_{\mathcal{A}}$ infinitely many times.

On the other hand, we start by showing that the Scott ranks of the tree of tuples of a structure and the Scott rank of the structure are equal. Towards this we need a few lemmas.

Theorem 0.3. *For any structure $\mathcal{A}$, $SR(\mathcal{A}) = SR(\mathcal{T}_{\mathcal{A}})$.*

Lemma 0.4. *Let $\mathcal{A}$ be a structure in vocabulary τ.*

(1) *Given a Π^{in}_{α} τ-formula φ, there is a Π^{in}_{α} formula T_{φ} in the vocabulary of labeled trees such that for any $\bar{a} \in \mathcal{A}$ and every $\sigma \in \mathcal{T}_{\mathcal{A}}$ with $\sigma \succeq \bar{a}$*

$$\mathcal{A} \models \varphi(\bar{a}) \iff \mathcal{T}_{\mathcal{A}} \models T_{\varphi}(\sigma).$$

(2) *Given a Π^{in}_{α} formula ψ in the vocabulary of labeled trees, there is a Π^{in}_{α} τ-formula ψ^* such that for any tuple $\bar{\sigma} = (\bar{a}_1, \dots, \bar{a}_n) \in \mathcal{T}_{\mathcal{A}}$*

$$\mathcal{T}_{\mathcal{A}} \models \psi(\bar{\sigma}) \iff \mathcal{A} \models \psi^*(\bar{a}, \dots, \bar{a}_n).$$

Proof. We first show Item 1 starting with quantifier free formulas. Let φ be quantifier free with free variables $x_1, \dots x_n$ in disjunctive normal form. For each disjunct φ_i let n_i be the index of the largest occuring variable and let T_{φ_i} be the formula expressing that the height of x in the tree is at least n_i and that $\varphi_i \subseteq l(x)$. Then $T_{\varphi} = \bigvee T_{\varphi_i}$. The construction of $\mathcal{T}_{\mathcal{A}}$ guarantees that for $\sigma \succeq \bar{a} \in \mathcal{A}$, $\mathcal{T}_{\mathcal{A}} \models T_{\varphi}(\sigma)$ if and only if $\mathcal{A} \models \varphi(\bar{a})$. Now, assume that we have a translation for $\varphi(\bar{x}, y)$ into $T_{\varphi}(x)$. If $\psi = \exists y \varphi(\bar{x}, y)$, then let

$$T_{\psi} = (\exists y \succeq x)\, T_{\varphi}(y).$$

We have that $\mathcal{A} \models \exists y\, \psi(\bar{a}, y)$, if and only if there is $b \in \mathcal{A}$ such that $\mathcal{A} \models \psi(\bar{a}, b)$ if and only if there is $\sigma \succeq \bar{a} \in \mathcal{T}_{\mathcal{A}}$ $\mathcal{T}_{\mathcal{A}} \models T_{\psi}(\sigma)$ if and only if $\mathcal{T}_{\mathcal{A}} \models T_{\psi}$. This proves the successor case. The limit case is immediate.

We now prove Item 2 by defining a family of Turing-computable embeddings from the class of structures containing $\mathcal{A}$ with parameters to labeled trees with parameters. Each of these embeddings will in some sense extend the Friedman-Stanley embedding that acts without parameters. Fix a tuple $\bar{\sigma} = (\bar{a}_1, \cdots, \bar{a}_n) \in \mathcal{T}_{\mathcal{A}}$. Consider any structure $\mathcal{B}$ in the class of $\mathcal{A}$ with parameters $\bar{p}$ such that $|\bar{p}| = |\bar{a}_1 \dots \bar{a}_n|$. We now construct the embedding $\Psi_{\bar{\sigma}}$. Non-uniformly fix a partition $\bar{p} = \bar{p}_1, \dots \bar{p}_n$ with $|\bar{p}_i| = |\bar{a}_i|$ and construct the tree of tuples of $\mathcal{B}$ as normal. To decide what elements become parameters in $\mathcal{T}_{\mathcal{B}}$, look for the lexicographical least elements τ_i coding the $\bar{p}_i$ such that the finite substructure of $\mathcal{T}_{\mathcal{B}}$ described by τ is isomorphic to the one described by σ in $\mathcal{T}_{\mathcal{A}}$. It is not difficult to check that this is a Turing computable embedding. Now, for any formula ψ in the language of labeled trees, any $\mathcal{B}$ in the class of $\mathcal{A}$ and any $\bar{\sigma} = (\bar{b}_0 \dots \bar{b}_n) \in \mathcal{T}_{\mathcal{B}}$ by the pull-back theorem for $\Psi_{\bar{\sigma}}$ there is ψ^* of the same complexity as ψ such that $\mathcal{B} \models \psi^*(\bar{b}_0 \dots \bar{b}_n)$ if and only if $\mathcal{T}_{\mathcal{B}} \models \psi(\bar{\sigma})$. As the image of $\Psi_{\bar{\sigma}}$ contains the parameterized tree $(\mathcal{T}_{\mathcal{A}}, \bar{\sigma})$, this proves Item 2. □

Proof. Let $\mathcal{A}$ be a structure such that $SR(\mathcal{T}_{\mathcal{A}}) = \gamma$. Consider a tuple $\bar{a} \in \mathcal{A}$. Let $\psi \in \Sigma^{\text{in}}_{\gamma}$ define the orbit of $\bar{a} \in \mathcal{T}_{\mathcal{A}}$. By Item 2 in Lemma 0.4 we have for all $\bar{b} \in \mathcal{A}$

$$\mathcal{A} \models \psi^*(\bar{b}) \iff \mathcal{T}_{\mathcal{A}} \models \psi(\bar{b}) \iff \bar{b} \cong \bar{a}.$$

Therefore, ψ^* defines the orbit of $\bar{a}$ as desired and hence, $SR(\mathcal{A}) \leq SR(\mathcal{T}_{\mathcal{A}})$.

We now show $SR(\mathcal{T}_{\mathcal{A}}) \leq SR(\mathcal{A}) = \gamma$. Let $\bar{\sigma} = \bar{a}_1 \dots \bar{a}_n$ be a tuple in $\mathcal{T}_{\mathcal{A}}$. The automorphism orbit of each $\bar{a}_i$ is $\Sigma^{\text{in}}_{\gamma}$ definable in $\mathcal{A}$, thus by Item 1 in Lemma 0.4

there is a $\Sigma^{\text{in}}_\gamma$ formula, χ_i, true only of elements τ extending σ that code elements automorphic to $\bar{a}_i$ in $\mathcal{A}$. Now, let ψ describe the finite subtree containing precisely the elements of $\bar{\sigma}$ as its leaves and let φ by the conjunction of ψ together with the χ_i's. Note that because the labels encode the height of the labeled element in the tree, ψ can be taken Σ^{in}_1. Let $\bar{\tau} = \bar{b}_1 \dots \bar{b}_n$ satisfy φ. Then, for every $\bar{b}_i$, the subtrees $T_{\bar{b}_i}$ rooted at $\bar{b}_i$ are isomorphic to the subtrees of $\bar{a}_i$ and as the tree $\mathcal{T}_\mathcal{A}$ is replicated we can extend these isomorphisms to an automorphism mapping $\bar{\sigma} \to \bar{\tau}$. □